SLURM-Only Memory Enforcement for Tactix Experiments

Tactix Development Project

27 August 2026

Outcome

The maintained benchmark infrastructure no longer uses `RLIMIT_AS`. SLURM's task cgroup is now the sole memory-enforcement authority. Allocation is expressed as 8 GiB per physical core; on the current SMT-2 nodes a strict 16-GiB task therefore requests two physical/four logical CPUs, while a targeted timing task requests six physical/twelve logical CPUs and exactly 48,000 MiB. No solver semantics changed.

Per project-owner direction, the current rank-1–40 campaign will be preserved as evidence under its original dual-limit contract. A fresh 80-arm screen will then rerun the same accepted Tactix and current-Kissat pins with no address-space limit and will become the selection authority.

1 Why the old jobs failed below 16 GiB RSS

Campaign `20260826-2340-tierd-ranks1-40-current-screen` requested `-mem=48000M` from SLURM, but its copied runner separately imposed `RLIMIT_AS=16 GiB`. Tactix ranks 5, 13, and 14 reported `std::bad_alloc` at peak RSS values of 13,970,768, 14,189,420, and 13,342,516 KiB. These were not 48,000-MiB cgroup OOM events.

An address-space limit counts virtual mappings. A capacity-growing vector or hash table may allocate replacement storage while its old storage remains live; that request can exceed the virtual-space ceiling even though resident high-water is much lower. The old screen therefore cannot support a memory-competitiveness decision.

2 Cluster enforcement evidence

An authoritative `scontrol show config` query at `2026-08-27T00:50:26-04:00` reported:

- `ProctrackType=proctrack/cgroup` and `TaskPlugin=task/cgroup`;
- `SelectTypeParameters=CR_CPU_MEMORY`; and
- `ConstrainRAMSpace=yes`, `AllowedRAMSpace=100.0\%`, and `AllowedSwapSpace=0.0\%`.

Consequently, `-mem` is a hard resident-memory task-cgroup ceiling and swap cannot hide an overage.

3 Infrastructure and policy changes

- `tools/bench/run.py` retains its CPU-time backstop but installs no memory rlimit. New SLURM manifests record `memory_enforcement=slurm_task_cgroup`; local diagnostics explicitly record that they have no memory enforcement.

- The frozen-baseline, Tactix Tier-D, LymphoSAT-attribution, and Phase-4 diagnostic generators now request total SLURM memory and two logical CPUs per physical core. Their maintained execution paths install no address-space cap.
- `DEVELOPMENT_PLAN.md` now states the physical-core rule and makes the strict 16-GiB shape explicit. It also separates a 48,000-MiB timing run from a later strict 16-GiB confirmation.
- Frozen campaigns and their copied runners remain immutable. Legacy manifest fields are read only for historical interpretation.

4 Validation

All 76 Release CTest tests pass. The new memory-policy regression parses each maintained runner and rejects any `setrlimit` call carrying `RLIMIT_AS`; it also guards the SMT-aware allocation formula and plan language. Campaign cleanup, Tier-D pinning, and Phase-4 allocation tests pass, as do Python compilation and all applicable documentation authority checks. The documentation size checker retains two pre-existing warning-level debts and reports no error.

No fuzz campaign is required for this increment because neither solver code nor proof/model semantics changed.

5 Next action

After this isolated infrastructure increment is committed and pushed, freeze a new rank-1–40 screen with the same accepted solvers and immutable inputs. Run 80 independent proof-producing solver tasks at `-c 12`, exactly `-mem=48000M`, 1,200 seconds, no throttle, no node constraint, and no `RLIMIT_AS`; independently validate every model and proof and require zero certificate residue. Use its complete aggregate to choose the next gap. A genuine Tactix memory solve loss has priority; otherwise proceed to Rank 4 and then Rank 32. Confirm any decision-changing 16-GiB claim separately at `-c 4/-mem=16G`.